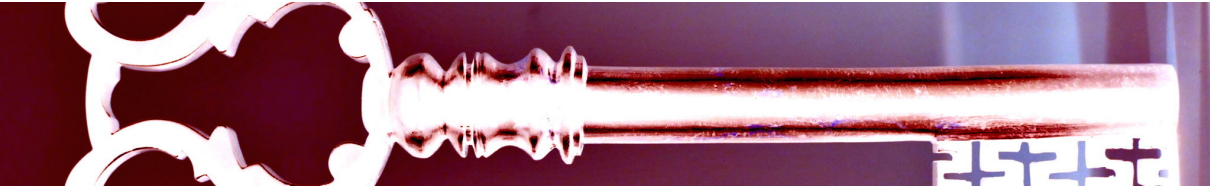
universität innsbruck



Number Theory

Cryptography

Rainer Böhme

23 April 2026

Agenda

1. **Modular arithmetic**
2. Extended Euclidean algorithm (EEA)
3. Primitive roots
4. Square-and-multiply algorithm (S&M)
5. Chinese remainder algorithm (CRA)

Basic Notions

All numbers are positive integers ($\in \mathbb{Z}^+$):

- $a \mid b \quad \Leftrightarrow \quad \exists k$ so that $k \cdot a = b$ — "divides"
- $a \equiv b \mod n \quad \Leftrightarrow \quad n \mid (a - b)$ — "congruent modulo n"
- $g \mid a \;\wedge\; g \mid b, \;\; \nexists k > g : k \mid a \;\wedge\; k \mid b \quad \Leftrightarrow \quad g = \gcd(a, b)$ — "greatest common divisor"
- $\gcd(a, b) = 1 \quad \Leftrightarrow \quad a$ and b are <u>coprimes</u>
- $\forall k < p : \gcd(p, k) = 1 \quad \Leftrightarrow \quad p$ is <u>prime</u> — $\neg$ "composite"
- $n = p \cdot q$ with p, q prime $\quad \Leftrightarrow \quad n$ is <u>semiprime</u>

Alternative notations:

$a \equiv b \mod n \quad \Leftrightarrow \quad a \equiv b \pmod n \quad \Leftrightarrow \quad a \equiv_n b \quad \Leftrightarrow \quad [a \mod n] = [b \mod n]$

Foundations

Euclidean division

$\forall a, b \quad \exists\, k, r$ such that $a = k \cdot b + r$ with $0 \leq r < b$.

r is called the remainder of the Euclidean division of a by b.

Bézout's identity

$\forall a, b \quad g = \gcd(a, b)$ always exists and $\exists\, u, v \in \mathbb{Z}$ such that

$$g = u \cdot a + v \cdot b.$$

Prime factorization

Every integer a can be written as a product of prime numbers

$$a = \prod_i p_i, \qquad p_i \text{ prime numbers}$$

This factorization is unique up to the order of the factors.

(These results generalize to the whole set of integers $\mathbb{Z}$.)

Euclid's Algorithm

Problem: Find the GCD of two integers a and b, $a > b$.

Idea: $a = k \cdot b + r$ where $k, r \in \mathbb{Z}^+ \Leftrightarrow \gcd(a, b) = \gcd(b, r)$

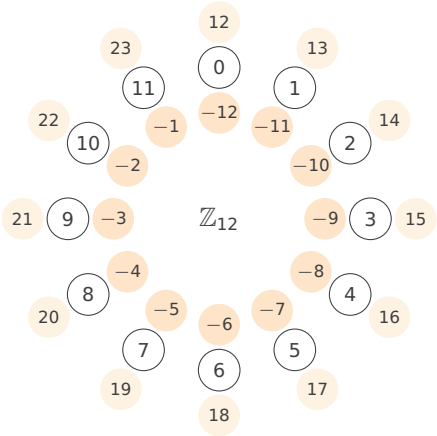
$$r \mid b \Rightarrow \gcd(b, r) = r$$

Example: $a = 2233$, $b = 399$

$a =$	$k \cdot$	$b +$	r
$2233 =$	$5 \cdot$	$399 +$	238
$399 =$	$1 \cdot$	$238 +$	161
$238 =$	$1 \cdot$	$161 +$	77
$161 =$	$2 \cdot$	$77 +$	7
$77 =$	$11 \cdot$	$\underline{7} +$	0

Modular Arithmetic

Let $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$. Each element of $\mathbb{Z}_n$ represents a congruence class modulo n.



Using the congruence relation ($\equiv$), we can calculate on this commutative ring of integers.

Addition: $5 + 8 = 13 \equiv 1 \mod 12$

Subtraction: $5 - 8 = -3 \equiv 9 \mod 12$

Multiplication by repeated addition

Exponentiation by repeated multiplication

Modular Division

- Based on multiplication:

$$a : b \equiv k \mod n \iff k \cdot b \equiv a \mod n$$

Examples in $\mathbb{Z}_{13}$:

$$2 \cdot 5 \equiv 10 \mod 13 \Rightarrow 10 : 5 \equiv 2 \mod 13$$

$$8 \cdot 10 \equiv 2 \mod 13 \Rightarrow 2 : 10 \equiv 8 \mod 13$$

- Defined only when the <u>multiplicative inverse</u> of b modulo n exists:

$$b^{-1} \in \mathbb{Z}_n \text{ such that } b \cdot b^{-1} \equiv 1 \mod n$$

- This does not always happen. (We'll see that later.)

Agenda

1. Modular arithmetic
2. **Extended Euclidean algorithm (EEA)**
3. Primitive roots
4. Square-and-multiply algorithm (S&M)
5. Chinese remainder algorithm (CRA)

Extended Euclidian Algorithm

Finding the multiplicative inverse of $a \in \mathbb{Z}_n$ when n is prime:

$$\gcd(a, n) = 1 \qquad \text{(n is coprime with all a)}$$

$$a \cdot u + n \cdot v = 1 \qquad \text{(Bézout's identity)}$$

$$a \cdot u = 1 - n \cdot v \equiv 1 \mod n \qquad \text{(by the definition of congruence)}$$

$$\rightarrow u \text{ is the multiplicative inverse of } a \qquad \text{(u and v can be found using the EEA)}$$

Extended Euclidian Algorithm

Problem: Given $a, b \in \mathbb{Z}$ with $\gcd(a, b) = 1$, find $u, v \in \mathbb{Z}$ so that

$$1 = a \cdot u + b \cdot v.$$

1. Euclid's algorithm

$$\begin{aligned}
a_0 &= k_0 \cdot b_0 + r_0 \\
a_1 &= k_1 \cdot b_1 + r_1 \\
\vdots &= \\
a_{i-1} &= k_{i-1} \cdot b_{i-1} + r_{i-1} \\
a_i &= k_i \cdot b_i + 1
\end{aligned}$$

2. Back-substitution and simplification

$$\begin{aligned}
1 &= u_1 \cdot b_0 + v_1 (a_0 - k_0 \cdot b_0) &&= u_0 \cdot a_0 + v_0 \cdot b_0 \\
1 &= u_2 \cdot (a_1 - k_1 \cdot a_2) + v_2 \cdot a_2 &&= u_1 \cdot a_1 + v_1 \cdot a_2 \\
\vdots &= && = \\
1 &= a_i - k_i \cdot (a_{i-1} - k_{i-1} \cdot a_i) &&= u_i \cdot a_{i-1} + v_i \cdot a_i \\
1 &= a_i - k_i \cdot b_i
\end{aligned}$$

Remember: $a_i = b_{i-1}, b_i = r_{i-1} \Rightarrow a_i = r_{i-2}$

Example

Find $x = 5^{-1}$ in $\mathbb{Z}_{13}$.

Euclid's algorithm

$$13 = 2 \cdot 5 + 3 \tag{1}$$
$$5 = 1 \cdot 3 + 2 \tag{2}$$
$$3 = 1 \cdot 2 + 1 \tag{3}$$

Back-substitution ... ... and simplification

$$1 = 3 - 1 \cdot 2 \tag{4}$$
$$1 = 3 - 1 \cdot (5 - 1 \cdot 3) \qquad = 2 \cdot 3 - 1 \cdot 5 \tag{5}$$
$$1 = 2 \cdot (13 - 2 \cdot 5) - 1 \cdot 5 \qquad = 2 \cdot 13 - 5 \cdot 5 \tag{6}$$

Solution: $x = -5 \equiv 8 \mod 13$

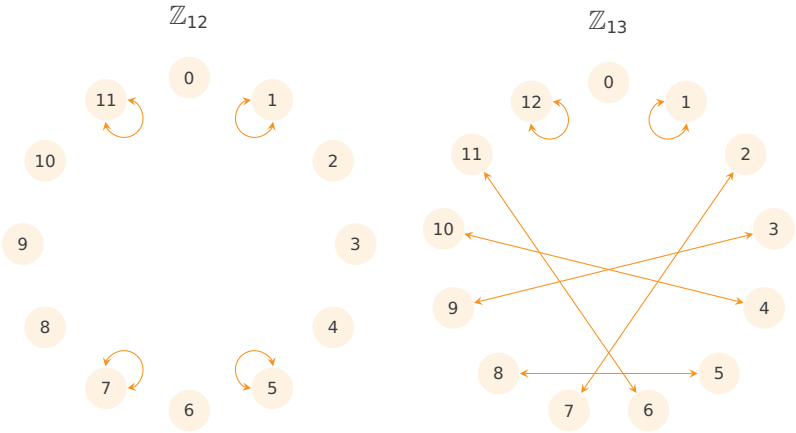
Second Example

Find $x = 18^{-1}$ in $\mathbb{Z}_{23}$.

$$
\begin{aligned}
23 &= 1 \cdot 18 + 5 && && (1)\\
18 &= 3 \cdot 5 + 3 && && (2)\\
5 &= 1 \cdot 3 + 2 && && (3)\\
3 &= 1 \cdot 2 + 1 && && (4)\\
1 &= -1 \cdot 2 + 1 \cdot 3 && && (5)\\
1 &= -1 \cdot (5 - 1 \cdot 3) + 1 \cdot 3 && &= -1 \cdot 5 + 2 \cdot 3 && (6)\\
1 &= -1 \cdot 5 + 2 \cdot (18 - 3 \cdot 5) && &= -7 \cdot 5 + 2 \cdot 18 && (7)\\
1 &= -7 \cdot (23 - 1 \cdot 18) + 2 \cdot 18 && &= -7 \cdot 23 + 9 \cdot 18 && (8)
\end{aligned}
$$

Check: $9 \cdot 18 = 162 \equiv 1 \mod 23$

Existence of Multiplicative Inverses



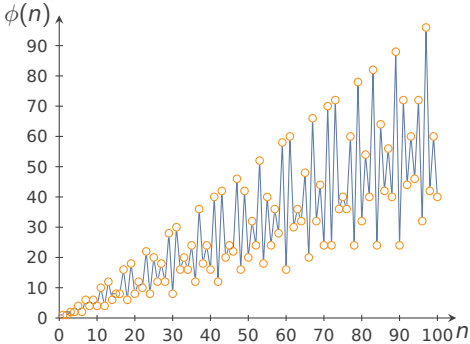
$$\exists x^{-1} \in \mathbb{Z}_n : x \cdot x^{-1} \equiv 1 \mod n \quad \Leftrightarrow \quad \gcd(n, x) = 1$$

Euler's Phi Function

$$\phi(n) = |\mathbb{Z}_n^*| = |\{1 \leq x < n \mid \gcd(n, x) = 1\}|$$

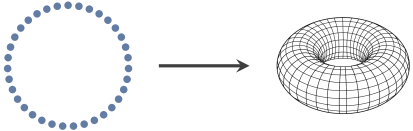
Lemmas useful to calculate ϕ

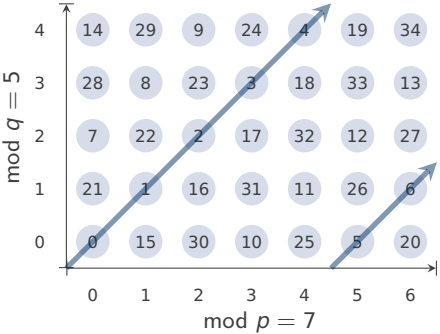
1. $\phi(p) = p - 1 \;\Leftrightarrow\; p$ prime
2. If $\gcd(p, q) = 1$, then $\phi(p \cdot q) = \phi(p) \cdot \phi(q)$.



The Donut Visualization

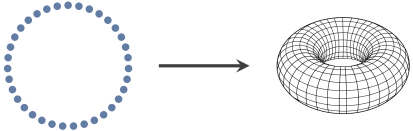
Mapping a semi-prime ring onto the surface of a torus.

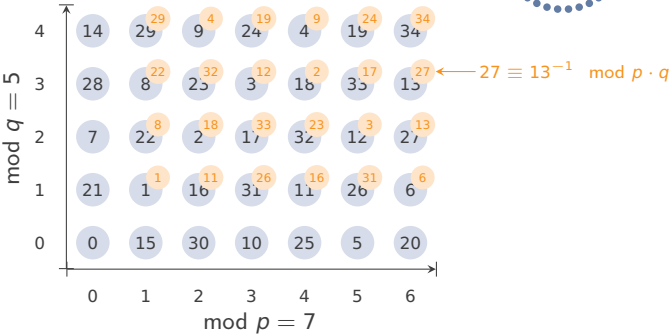




The Donut Visualization

Mapping a semi-prime ring onto the surface of a torus.





Finite Fields $\mathbb{Z}_p$

A field $\mathbb{F}$ is an algebraic structure with the following properties:

- **Closure** $\forall a, b \in \mathbb{F} : a + b \in \mathbb{F} \wedge a \cdot b \in \mathbb{F}$
- **Associativity**
  $a + (b + c) = (a + b) + c \quad \wedge \quad a \cdot (b \cdot c) = (a \cdot b) \cdot c$
- **Commutativity** $a + b = b + a \quad \wedge \quad a \cdot b = b \cdot a$
- **Identity** $a + 0 = a \quad \wedge \quad a \cdot 1 = a$
- **Inverses** $a + (-a) = 0 \quad \wedge \quad a \cdot a^{-1} = 1 \; \forall a \in \mathbb{F} \setminus \{0\}$
- **Distributivity** $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

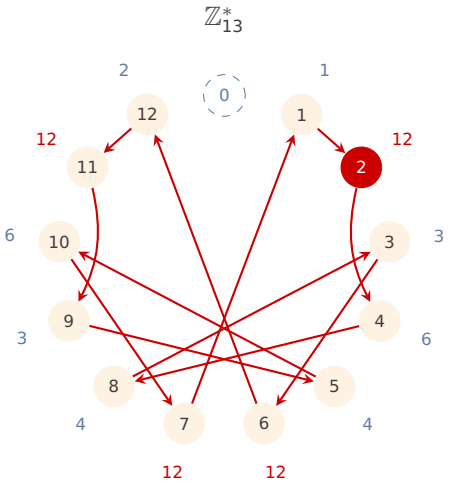
A finite field is a field with finitely many elements.

For all primes $p > 1$, $\mathbb{Z}_p = \mathbb{Z}_p^* \cup \{0\}$ is a finite field.

Agenda

1. Modular arithmetic
2. Extended Euclidean algorithm (EEA)
3. **Primitive roots**
4. Square-and-multiply algorithm (S&M)
5. Chinese remainder algorithm (CRA)

Modular Powers



$$\left|\{a^1 \bmod p, a^2 \bmod p, \ldots, a^{p-1} \bmod p\}\right|$$

Definitions

Primitive root

$g \in \mathbb{Z}_p^*$ is primitive root modulo prime p if for any $a \in \{1, \ldots, p-1\}$ $\exists k$ such that $a \equiv g^k \mod p$.

Generator

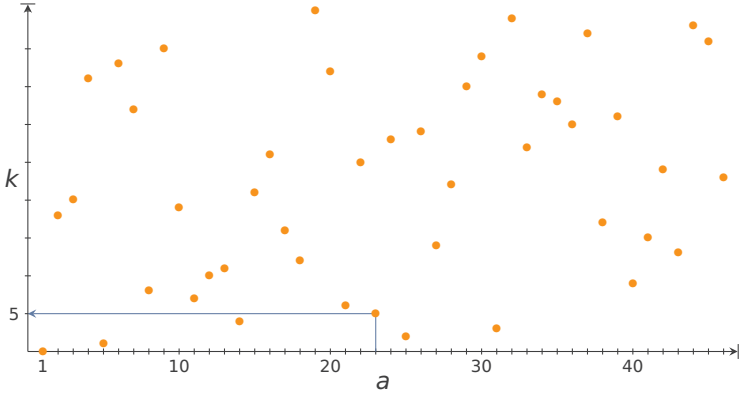
g is generator of the multiplicative group of integers modulo p.

Discrete logarithm

k is the discrete logarithm of a to the base g modulo p.

Discrete Logarithm

$$a \equiv g^k \mod p \quad \text{with } p = 47 \text{ and } g = 5$$



Fermat's Little Theorem

For any integer a and prime p:

$$p \mid (a^p - a)$$

Alternative forms in modular arithmetic:

$$a^p \equiv a \mod p$$

$$a^{p-1} \equiv 1 \mod p$$

Generalization to <u>Euler's theorem</u> if $\gcd(a, n) = 1$:

$$a^{\phi(n)} \equiv 1 \mod n$$

Agenda

1. Modular arithmetic
2. Extended Euclidean algorithm (EEA)
3. Primitive roots
4. **Square-and-multiply algorithm (S&M)**
5. Chinese remainder algorithm (CRA)

Square and Multiply

Calculate $x \equiv 22^{85} \mod 37$.

$$85 = 1010101_2$$

$$22 \xrightarrow[\uparrow 2]{} 22^2 \xrightarrow[\uparrow 2]{} 22^4 \xrightarrow[\times 22]{} 22^5 \xrightarrow[\uparrow 2]{} 22^{10} \xrightarrow[\uparrow 2]{} 22^{20} \xrightarrow[\times 22]{} 22^{21} \xrightarrow[\uparrow 2]{} 22^{42} \xrightarrow[\uparrow 2]{} 22^{84} \xrightarrow[\times 22]{} 22^{85}$$

$$22^2 = 484 \equiv 3 \mod 37 \quad (1)$$

$$3^2 \Rightarrow 9 \quad \times\ 22 = 198 \equiv 13 \mod 37 \quad (2)$$

$$13^2 = 169 \equiv 21 \mod 37 \quad (3)$$

$$21^2 = 441 \equiv 34 \Rightarrow (-3) \quad \times\ 22 = (-66) \equiv 8 \mod 37 \quad (4)$$

$$8^2 = 64 \equiv 27 \mod 37 \quad (5)$$

$$27^2 \equiv (-10)^2 = 100 \Rightarrow (-11) \quad \times\ 22 = (-242) \equiv \underline{17} \mod 37 \quad (6)$$

Agenda

1. Modular arithmetic
2. Extended Euclidean algorithm (EEA)
3. Primitive roots
4. Square-and-multiply algorithm (S&M)
5. **Chinese remainder algorithm (CRA)**

Chinese Remainder Theorem

Chinese Remainder Theorem (CRT)

Relation between $\mathbb{Z}_n$ and $\mathbb{Z}_{p_1}, \mathbb{Z}_{p_2}, \ldots$ with $n = \prod_i p_i$ and $\gcd(p_i, p_j) = 1 \; \forall (i,j)$.

For any $y_1, y_2, \ldots,$ there exists an integer a such that

$$a \equiv y_i \bmod p_i \quad \wedge \quad a \equiv y_j \bmod p_j \quad \wedge \ldots$$

For any solution b it holds:

$$\begin{array}{ccccccc} a \equiv b \bmod n & \Leftrightarrow & a \equiv b \bmod p_i & \wedge & a \equiv b \bmod p_j & \wedge & \ldots \\ \text{because} \updownarrow & & \updownarrow & & \updownarrow & & \\ n \mid (a-b) & \overset{\gcd(p_i,p_j)=1}{\Leftrightarrow} & p_i \mid (a-b) & \wedge & p_j \mid (a-b) & \wedge & \ldots \end{array}$$

Chinese Remainder Algorithm (CRA)

1. Calculate $y_i \in \mathbb{Z}_{p_i} \equiv f(x) \mod p_i$ for all p_i.
2. Combine $y = \sum_i y_i \frac{n}{p_i} \left[\left(\frac{n}{p_i} \right)^{-1} \right]_{p_i}$ multiplicative inverse in $\mathbb{Z}_{p_i}$

Proof Idea for the CRA

Special case: $n = p \cdot q$

$$1 = u \cdot p + v \cdot q$$

Bézout's identity

$$a = (u \cdot p) \cdot y_q + (v \cdot q) \cdot y_p$$

$$\begin{cases} \equiv y_p & \bmod p \\ \equiv y_q & \bmod q \end{cases}$$

Because:

	mod p	mod q
$u \cdot p$	0	1
$v \cdot q$	1	0
a	$0 \cdot y_q + 1 \cdot y_p$	$1 \cdot y_q + 0 \cdot y_p$

Example

$$a \equiv 2 \mod p \qquad p = 11$$
$$a \equiv 6 \mod q \qquad q = 13$$

Find a solving the system with the CRA.

$$1 = 6 \cdot 11 \quad + (-5) \cdot 13 \qquad \text{(EEA)}$$
$$a = 6 \cdot 11 \cdot 6 + (-5) \cdot 13 \cdot 2 \equiv \underline{123} \mod n$$

Any other integer equivalent to 123 modulo $11 \cdot 13 = 143$ is a solution, and vice versa.

Further Reading

- J. H. Silverman (2005): A Friendly Introduction to Number Theory, 3rd edition, Prentice Hall, 493 pages.

Syllabus

Summer Term 2026

	05.03.26	1.	Introduction, basic terminology	
	12.03.26	2.	Probability, information, and coding theory for cryptography	
	19.03.26	3.	Symmetric encryption, perfect secrecy	
Tue	24.03.26	4.	Block ciphers	HSB 9
	16.04.26	5.	Cryptanalysis of classical ciphers	
	23.04.26	6.	Number theory for cryptography	
	30.04.26	7.	Computational complexity for cryptography	
	07.05.26	8.	Asymmetric encryption	
Tue	19.05.26	9.	Message authentication	HSB 9
	28.05.26	10.	Elliptic curve cryptography	
	11.06.26	11.	Commitments, zero-knowledge, and secret sharing	
Tue	23.06.26	12.	Q & A for the exam	HSB 9
	25.06.26		**Written exam**	

Last update: April 23, 2026